YASSINE ABASSI

Montreal, QC · (514) 953-7120 · yassine020528@gmail.com · linkedin.com/in/yassine-abassi
yassineabassiportfolio.netlify.app

SUMMARY

Computer engineering student at Polytechnique Montréal with hands-on experience building full-stack data platforms, real-time systems, and AI-driven applications. Strong foundation in Python, Angular, .NET, and data pipeline design. Passionate about applying computer engineering to solve complex problems in quantitative finance and investment technology.

EDUCATION

B.Eng. Computer Engineering, Polytechnique Montréal — Aug 2022 – May 2026
120 credits (105 completed, 15 in progress)

EXPERIENCE

Software Quality Assurance Intern — May 2024 – Sep 2024
UPTOTEST · Ariana, Tunisia

- Designed and automated end-to-end functional test suites for the KOORS web app using Cypress with Cucumber/Gherkin in a BDD workflow.
- Translated product requirements into testable user stories and acceptance criteria, managed defects and traceability in Jira across sprints.
- Participated in Agile Scrum ceremonies to align QA priorities with development milestones and release timelines.

TECHNICAL SKILLS

Languages	C, C++, C#, Java, Kotlin, Python, JavaScript, TypeScript, VHDL, Assembly
Web & Mobile	Angular, React.js, Node.js, NestJS, Express.js, Jetpack Compose, .NET / ASP.NET Core, Three.js
Data & APIs	PostgreSQL, MongoDB, SQLite, Room, Entity Framework Core, Retrofit, OkHttp, GDAL, RxJS
DevOps & Tools	Git, GitLab, Docker, Jira, Swagger, Cypress, Jest, Jasmine, Mocha, Android Studio, Vivado, Wireshark
Robots & Boards	Zynq7000, ATmega164A, Pynq-Z1, Basys3 Artix-7, AgileX Limo

PROJECTS

Hydro-Québec Weather Forecasting Platform | *Angular 19 · .NET 8 · EF Core · GDAL · Leaflet · Chart.js*

- Built a full-stack weather visualization platform with Angular 19 SSR and ASP.NET Core 8 Web API for real-time multi-model forecast comparison with interactive geospatial mapping.
- Implemented GDAL raster processing, Leaflet cartographic rendering, Chart.js dashboards, and Handsontable data grids; designed RESTful APIs with Swagger and Repository/UoW patterns.

Android Mobile Applications (3 apps) | *Kotlin · Jetpack Compose · Room · Google Maps SDK · Retrofit*

- Finance tracker with MVVM, Room persistence, and Material 3 Adaptive layouts; Bluetooth device tracker with GPS trajectories, Maps SDK, and Coroutines; location-based running-walking app with Retrofit/OkHttp networking and Compose Navigation.

Multi-Robot Exploration System | *Python · ROS2 · Gazebo · Docker*

- Co-designed an autonomous multi-robot exploration system with AgileX Limo robots, implementing navigation, sensor fusion, and inter-robot coordination in containerized modules.

Real-Time Multiplayer Combat Game | *Angular · NestJS · WebSockets*

- Developed a real-time multiplayer game with bidirectional WebSocket communication, matchmaking, event synchronization, and responsive UI under concurrent loads.

Embedded Obstacle Detection System | *C · ATmega164A*

- Built an autonomous obstacle detection and search system with low-level I/O control, sensor acquisition, and real-time decision logic on bare-metal microcontroller.

3D Interactive Portfolio | *React.js · Three.js*

- Crafted a 3D portfolio with an interactive room featuring a functional OS, custom camera rigging, and state-driven transitions across mobile and desktop.

ADDITIONAL

- **PolyHacks 2025** — Built WildGuard, an AI ecological monitoring system using YOLOv8 for real-time wildlife recognition on live video.
- **Private Tutor** — Mathematics, Physics & CS for college/university students (2021 – Present).
- **Languages:** English, French, Arabic (fluent) | German (intermediate)